

PROVISIONAL PATENT APPLICATION

Under 35 U.S.C. § 111(b)

TITLE OF INVENTION: MODULAR WRISTBAND HEALTH MONITORING APPARATUS WITH ZERO-HOLE DUAL-CONNECTOR ARCHITECTURE, BIDIRECTIONAL NEUROMUSCULAR ELECTRODE ARRAY, AND INTEGRATED BREATH ANALYSIS CHANNEL

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Note: This application is filed by the inventor as an individual. The EoS Foundation is listed as the inventor's research affiliation. The inventor retains full ownership of this patent. No assignment has been made to any organization.

ENTITY STATUS: Small Entity (Individual Inventor)

RELATED APPLICATION: This application is related to U.S. Provisional Application No. 64/073,334, filed May 23, 2026, titled "HEALTH-KEY ULTRA: Secure Multi-Function USB-C Key with Zero-Hole Architecture and Integrated Health Monitoring," by the same inventor. The present application discloses a distinct wristband form factor embodiment with additional neuromuscular interface and breath analysis features not disclosed in the prior application.

FIELD OF THE INVENTION

The present invention relates to wearable health monitoring devices, and more particularly to a modular wristband apparatus integrating a zero-hole dual-connector architecture, a

bidirectional neuromuscular electrode array capable of both surface electromyography (sEMG) sensing and transcutaneous electrical nerve stimulation (TENS) therapy, and an integrated breath analysis channel for non-invasive blood alcohol concentration (BAC) and volatile organic compound (VOC) measurement.

BACKGROUND OF THE INVENTION

Wearable health monitoring devices have proliferated in recent years, with commercially available products including smartwatches, fitness bands, and medical-grade monitors. Despite this growth, existing devices suffer from fundamental architectural limitations that prevent integration of monitoring, diagnostics, and therapy in a single wrist-worn form factor.

First, conventional wristbands require dedicated charging ports, data ports, or both, creating physical openings in the device housing that compromise water resistance, structural integrity, and aesthetic design. Devices such as the Apple Watch Series 9, Fitbit Charge 6, and Garmin Vivosmart 5 all incorporate proprietary magnetic charging connectors or exposed port openings that represent structural vulnerabilities.

Second, no commercially available wristband device integrates both surface electromyography (sEMG) sensing and transcutaneous electrical nerve stimulation (TENS) therapy on a shared electrode array. Existing TENS wearables (PowerDot 2.0, Compex Mini Wireless) provide therapy output only, with no sensing capability. Existing sEMG gesture recognition bands (Myo Armband, Meta EMG prototype) provide sensing only, with no therapeutic output. The combination of bidirectional functionality on shared platinum electrodes in a wristband form factor has not been previously disclosed.

Third, wrist-worn breath analysis for blood alcohol concentration (BAC) measurement has been limited to transdermal alcohol sensing (BACtrack Skyn), which measures alcohol diffused through the skin rather than exhaled breath, resulting in significant measurement lag (30–90 minutes) and reduced accuracy compared to direct breath analysis. No prior art device integrates a Venturi-accelerated breath analysis channel with an electrochemical fuel cell and MOx VOC sensor array within a wristband clasp housing.

There exists a need for a wristband apparatus that: (a) eliminates all dedicated port openings by repurposing the mechanical clasp as the sole charging and data interface; (b) provides bidirectional neuromuscular functionality on shared skin-contact electrodes; and

© integrates a breath analysis channel within the clasp housing for real-time BAC and VOC measurement.

SUMMARY OF THE INVENTION

The present invention provides a modular wristband health monitoring apparatus comprising three primary novel subsystems:

First, a Zero-Hole Architecture in which the wristband clasp incorporates a USB Type-C female receptacle on one end and a USB Type-C male plug on the other end, such that the USB-C connectors serve simultaneously as the mechanical clasp fastener, the charging interface, the data interface, and the breath analysis inlet, with no additional openings in the device housing.

Second, a Bidirectional Neuromuscular Interface comprising a shared array of six platinum electrodes embedded in the flexible wristband strap, wherein the electrode array is selectively configurable by a multiplexer circuit to operate in a first mode as a surface electromyography (sEMG) sensor for gesture recognition and muscle activity monitoring, or in a second mode as a transcutaneous electrical nerve stimulation (TENS) therapy output for pain management and muscle rehabilitation.

Third, an Integrated Breath Analysis Channel disposed within the USB-C clasp housing, comprising a hexagonal titanium grille, a PTFE hydrophobic membrane, a Venturi constriction channel, an electrochemical fuel cell for BAC measurement, and a metal-oxide semiconductor (MOx) VOC sensor for ketosis and biomarker detection, wherein exhaled breath directed toward the USB-C female receptacle is channeled through the analysis pathway and exhausted through a side port.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated herein and constitute a part of this specification, illustrate preferred embodiments of the invention.

FIG. 1 is a system block diagram of the HEALTH-BAND Neuro, showing the Nordic nRF52840 MCU as the central processing unit connected to all sensor subsystems, the

BQ25185 PMIC, the USB-C dual-connector interface, the OLED display, and the NAND flash storage.

FIG. 2 is a longitudinal cross-section view of the Zero-Hole Architecture clasp detail, showing the breath analysis channel components including the USB-C female receptacle (101), hexagonal titanium grille (102), PTFE hydrophobic membrane (103), Venturi channel (104), Dart EC4-10-100 fuel cell (105), SGP40 VOC sensor (106), and USB-C male plug (201).

FIG. 3 is a cross-section view of the flexible wristband strap showing the bidirectional neuromuscular interface, including the six platinum electrodes (301a–301f), the multiplexer (304), the sEMG input path (302) comprising an instrumentation amplifier, DSP filter, and TinyML classifier, and the TENS output path (303) comprising a boost converter, H-bridge, and pulse generator.

FIG. 4 is a perspective view of the complete HEALTH-BAND Neuro assembly worn on the left wrist, showing the core module (401), flexible strap (402), USB-C female receptacle (403), USB-C male plug (404), OLED display (405), electrode contacts (406a–406f), and magnetic latch mechanism (407).

FIG. 5 is an exploded view of the modular Core-Strap Architecture, showing the Core Module (500) comprising the MCU PCB (501), PMIC (502), battery (503), NAND flash (504), USB-C female housing with Venturi channel (505), and USB-C male clasp plug (506), connected to the Flexible Strap Module (510) via the 14-pin FPC connector (520).

FIG. 6 is a longitudinal cross-section view and flow diagram of the Integrated Breath Analysis Channel, showing the complete breath pathway from inlet (601) through titanium grille (602), PTFE membrane (603), Venturi constriction (604), electrochemical fuel cell (605), VOC sensor (606), to exhaust port (607), with breath flow direction arrows (608).

FIG. 7 is a power and data flow diagram showing the Pass-Through Power and Data Architecture, including the host device (700), USB-C male plug (701), BQ25185 PMIC (702), Li-Po battery (703), Nordic nRF52840 MCU (704), USB-C female receptacle (705), downstream device (706), USB-PD power path, pass-through power delivery path, and BLE 5.3 wireless data path.

DETAILED DESCRIPTION OF THE PREFERRED

EMBODIMENTS

I. System Overview

Referring to FIG. 1 and FIG. 4, the HEALTH-BAND Neuro (400) is a modular wristband health monitoring apparatus comprising a Core Module (401) and a Flexible Strap Module (402). The Core Module houses the primary processing, power management, storage, and clasp interface electronics. The Flexible Strap Module houses the skin-contact sensors and electrode array. The two modules are connected via a 14-pin flexible printed circuit board (FPCB) connector (520) as shown in FIG. 5.

The central processing unit is a Nordic Semiconductor nRF52840 microcontroller (MCU) operating at 64 MHz with an ARM Cortex-M4 core, 1 MB Flash, 256 KB RAM, and an integrated BLE 5.3 radio. The MCU interfaces with all sensor subsystems via I²C, SPI, and UART buses.

II. Zero-Hole Architecture

Referring to FIG. 2 and FIG. 7, the Zero-Hole Architecture is the primary structural innovation of the present invention. The wristband clasp comprises two USB Type-C connectors disposed at opposing ends of the Core Module housing:

USB-C Female Receptacle (101/403/601/705): A standard USB Type-C female receptacle mounted at the left end of the Core Module housing. This receptacle serves three simultaneous functions: (a) mechanical clasp engagement with the USB-C male plug; (b) USB 2.0 and USB Power Delivery (USB-PD) electrical interface for charging and data; and © breath analysis inlet through which exhaled breath enters the integrated breath analysis channel.

USB-C Male Plug (201/404/506/701): A standard USB Type-C male plug mounted at the right end of the Core Module housing. This plug serves three simultaneous functions: (a) mechanical clasp engagement with a host device USB-C port or the female receptacle; (b) USB 2.0 and USB-PD electrical interface presenting the device as a composite USB device (HID + MSC); and © ECG electrode via the shield conductor of the USB-C cable.

The device housing contains no additional openings, ports, buttons, or gaps beyond the two USB-C connector faces. This architecture achieves IP67 water resistance rating without requiring separate port covers or seals.

Pass-Through Power Delivery (FIG. 7): When the USB-C male plug (701) is connected to a host device (700), the BQ25185 PMIC (702) negotiates USB-PD at 5V/3A and simultaneously: (a) charges the internal 200 mAh Li-Po battery (703); (b) powers the Nordic nRF52840 MCU (704); and © passes through up to 5V/2A to the USB-C female receptacle (705) for charging a downstream device (706) such as a smartphone. This pass-through capability allows the HEALTH-BAND Neuro to function as a USB hub while being worn.

III. Bidirectional Neuromuscular Interface

Referring to FIG. 3, the Flexible Strap Module (510) incorporates six platinum electrodes (301a–301f) embedded in the inner surface of the flexible strap at positions corresponding to the flexor digitorum superficialis, flexor carpi radialis, flexor carpi ulnaris, extensor digitorum, extensor carpi radialis, and extensor carpi ulnaris muscle groups of the forearm.

Electrode Composition: Each electrode is a 10 mm diameter platinum disc with a surface roughness of $R_a \leq 0.4 \mu\text{m}$, providing low contact impedance ($< 5 \text{ k}\Omega$ at 1 kHz) for sEMG acquisition and uniform current distribution for TENS delivery. Platinum is selected for its biocompatibility (ISO 10993 compliant), corrosion resistance, and electrochemical stability across the voltage range required for both sEMG sensing ($\pm 5 \text{ mV}$) and TENS delivery (0–80V peak).

Multiplexer Circuit (304): A CMOS analog multiplexer (ADG1606 or equivalent) controlled by the MCU GPIO selects the operational mode of the electrode array. In sEMG mode, the multiplexer connects the electrodes to the sEMG input path (302). In TENS mode, the multiplexer connects the electrodes to the TENS output path (303). The transition between modes requires a minimum 100 ms isolation period enforced by firmware to prevent cross-mode interference.

sEMG Input Path (302): In sEMG mode, differential electrode pairs are connected to an INA128 instrumentation amplifier (gain = 1000, CMRR > 100 dB) followed by a 20–500 Hz bandpass filter implemented in the MCU DSP. The filtered signal is processed by a TinyML gesture classifier (16 gesture vocabulary, < 50 ms inference latency) running on the MCU's ARM Cortex-M4 core with CMSIS-DSP acceleration.

TENS Output Path (303): In TENS mode, a boost converter raises the battery voltage to the required therapeutic voltage (0–80V peak, programmable). An H-bridge circuit generates biphasic symmetric waveforms with programmable pulse width (50–400 μs), frequency (1–150 Hz), and amplitude. Five clinical TENS protocols are stored in firmware:

conventional (80–150 Hz, 50–80 μ s), acupuncture-like (1–4 Hz, 200–400 μ s), burst (100 Hz bursts at 1–2 Hz), brief-intense (80–150 Hz, 150–250 μ s), and hyperstimulation (1–10 Hz, 250–400 μ s).

IV. Integrated Breath Analysis Channel

Referring to FIG. 6, the Integrated Breath Analysis Channel (600) is disposed within the USB-C clasp housing, utilizing the USB-C female receptacle opening (601) as the breath inlet.

Titanium Hexagonal Grille (602): A hexagonal-pattern titanium mesh with 150 μ m aperture diameter, positioned immediately behind the USB-C female receptacle face, filters particulates and provides structural support. Titanium is selected for biocompatibility and corrosion resistance.

PTFE Hydrophobic Membrane (603): A polytetrafluoroethylene (PTFE) membrane with 0.2 μ m pore size and > 98% water vapor rejection, positioned downstream of the titanium grille, prevents liquid water ingress while permitting passage of breath vapor and gaseous analytes.

Venturi Constriction Channel (604): A converging-diverging channel geometry with a throat diameter of 2.0 mm and a convergence half-angle of 15°, designed to accelerate breath flow to a minimum velocity of 2 m/s at the throat for a normal exhalation flow rate of 0.5 L/s. The Venturi geometry ensures consistent analyte delivery to the downstream sensors independent of exhalation force variation.

Electrochemical Fuel Cell (605): A Dart Sensors EC4-10-100 electrochemical fuel cell sensor for blood alcohol concentration (BAC) measurement. The sensor produces a current proportional to ethanol concentration in the range 0–0.40% BAC with an accuracy of $\pm 0.005\%$ BAC at concentrations below 0.10% BAC. The sensor response time is < 15 seconds for a 90% full-scale reading.

SGP40 MOx VOC Sensor (606): A Sensirion SGP40 metal-oxide semiconductor sensor for volatile organic compound (VOC) detection. The sensor provides a VOC index (1–500) correlated with acetone (ketosis biomarker), hydrogen sulfide (oral microbiome marker), and ammonia (kidney function marker) concentrations. The sensor is operated at 85°C heater temperature for optimal sensitivity.

Exhaust Port (607): A 1.0 mm diameter exhaust port on the lateral face of the clasp housing allows breath gas to exit after passing through the sensor array, preventing back-

pressure buildup that would impede exhalation.

V. Additional Sensor Suite

The HEALTH-BAND Neuro incorporates the following additional sensors in the Flexible Strap Module:

Photoplethysmography (PPG) / SpO2: Maxim Integrated MAX30101 optical sensor with 660 nm (red) and 880 nm (infrared) LEDs for heart rate (HR), heart rate variability (HRV), and blood oxygen saturation (SpO2) measurement. Sampling rate: 100 Hz. SpO2 accuracy: $\pm 2\%$ (70–100% range).

Electrocardiography (ECG): Texas Instruments OPA391 precision amplifier configured as a single-lead ECG front-end, with the ECG signal derived from the differential voltage between the USB-C male plug shield electrode (right wrist contact) and the platinum electrode array (left wrist contact). This configuration provides a Lead I equivalent ECG waveform.

Inertial Measurement Unit (IMU): Bosch BMI270 6-axis IMU (3-axis accelerometer + 3-axis gyroscope) for motion artifact detection, step counting, sleep staging, and tremor analysis. The BMI270 incorporates an on-chip gesture recognition engine for activity classification.

Skin Temperature: ams AS6221 digital temperature sensor with $\pm 0.09^\circ\text{C}$ accuracy for skin temperature monitoring and fever detection.

UV Index: Vishay VEML6075 UVA/UVB sensor for ultraviolet radiation exposure monitoring and sunburn risk assessment.

VOC / Ketosis: Sensirion SGP40 MOx sensor (shared with breath analysis channel) for ambient VOC monitoring when not in breath analysis mode.

VI. Data Storage and Connectivity

Storage: Winbond W25N512GV 64 GB NAND Flash provides local data storage for health records, firmware, gesture models, and TENS protocols. The MCU interfaces with the flash via SPI at 80 MHz.

Display: Solomon Systech SSD1306 0.49-inch OLED display (64×32 pixels) provides real-time health metric readout and device status indication.

Wireless: BLE 5.3 (integrated in Nordic nRF52840) provides wireless connectivity to the EoS Health companion application on iOS and Android. BLE advertising interval: 100 ms. Connection interval: 7.5–15 ms. Maximum throughput: 1.37 Mbps (2M PHY).

Wired: USB 2.0 Full Speed (12 Mbps) via USB-C male plug, presenting as a composite device: USB HID (health data streaming) + USB MSC (mass storage for health records export).

CLAIMS

The following claims define the scope of the invention. For a provisional application, these claims are illustrative and non-limiting; the claims of the corresponding non-provisional application may be broader or narrower.

Claim 1. A wristband health monitoring apparatus comprising: a core module housing having a first end and a second end; a USB Type-C female receptacle mounted at the first end of the core module housing; a USB Type-C male plug mounted at the second end of the core module housing, wherein the USB Type-C female receptacle and the USB Type-C male plug together constitute the sole mechanical clasp of the wristband and the sole electrical interface of the core module housing, such that the core module housing contains no additional openings, ports, or electrical connectors; a flexible strap connected to the core module housing; and a microcontroller disposed within the core module housing and electrically connected to both the USB Type-C female receptacle and the USB Type-C male plug.

Claim 2. The apparatus of Claim 1, wherein the USB Type-C female receptacle is configured to simultaneously: (a) mechanically engage the USB Type-C male plug to form the wristband clasp; (b) receive USB Power Delivery (USB-PD) electrical power from a host device connected to the USB Type-C male plug; and © receive exhaled breath from a user for breath analysis.

Claim 3. The apparatus of Claim 1, further comprising a breath analysis channel disposed within the core module housing between the USB Type-C female receptacle and an exhaust port, the breath analysis channel comprising: a hydrophobic membrane; a Venturi constriction; an electrochemical fuel cell sensor; and a metal-oxide semiconductor (MOx) volatile organic compound sensor.

Claim 4. The apparatus of Claim 3, wherein the electrochemical fuel cell sensor is configured to measure blood alcohol concentration (BAC) in exhaled breath with an accuracy of $\pm 0.005\%$ BAC or better.

Claim 5. The apparatus of Claim 3, wherein the MOx volatile organic compound sensor is configured to detect at least one of: acetone as a ketosis biomarker; hydrogen sulfide as an oral microbiome marker; or ammonia as a kidney function marker.

Claim 6. The apparatus of Claim 1, further comprising: a plurality of platinum electrodes embedded in an inner surface of the flexible strap; a multiplexer circuit electrically connected to the plurality of platinum electrodes and controlled by the microcontroller; a surface electromyography (sEMG) signal acquisition circuit; and a transcutaneous electrical nerve stimulation (TENS) signal generation circuit; wherein the multiplexer circuit is selectively configurable to connect the plurality of platinum electrodes to the sEMG signal acquisition circuit in a first mode, or to the TENS signal generation circuit in a second mode.

Claim 7. The apparatus of Claim 6, wherein the plurality of platinum electrodes comprises six platinum disc electrodes having a diameter of approximately 10 mm and a surface roughness of $R_a \leq 0.4 \mu\text{m}$.

Claim 8. The apparatus of Claim 6, wherein the sEMG signal acquisition circuit comprises an instrumentation amplifier, a bandpass filter, and a machine learning gesture classifier configured to recognize at least 16 distinct hand gestures.

Claim 9. The apparatus of Claim 6, wherein the TENS signal generation circuit comprises a boost converter, an H-bridge circuit, and a pulse generator configured to generate biphasic symmetric waveforms with programmable pulse width in the range of 50–400 microseconds and programmable frequency in the range of 1–150 Hz.

Claim 10. The apparatus of Claim 1, further comprising a power management integrated circuit configured to: receive USB Power Delivery electrical power from a host device via the USB Type-C male plug; charge an internal battery from the received power; power the microcontroller from the received power or the internal battery; and simultaneously provide pass-through power to a downstream device connected to the USB Type-C female receptacle.

Claim 11. The apparatus of Claim 1, wherein the flexible strap further comprises: a photoplethysmography sensor for heart rate and blood oxygen saturation measurement;

an electrocardiography amplifier circuit; a skin temperature sensor; and an inertial measurement unit.

Claim 12. The apparatus of Claim 11, wherein the electrocardiography amplifier circuit is configured to derive an electrocardiography signal from a differential voltage between the USB Type-C male plug shield conductor and at least one of the plurality of platinum electrodes.

Claim 13. A method of simultaneously providing mechanical fastening, electrical charging, and breath analysis in a wristband device, the method comprising: engaging a USB Type-C male plug disposed at a first end of a wristband core module with a USB Type-C female receptacle disposed at a second end of the wristband core module to mechanically fasten the wristband; receiving electrical power via the USB Type-C male plug from a host device; and directing exhaled breath through the USB Type-C female receptacle into a breath analysis channel disposed within the core module to measure at least one of blood alcohol concentration or volatile organic compound concentration.

Claim 14. The method of Claim 13, further comprising passing through electrical power from the USB Type-C male plug to the USB Type-C female receptacle to charge a downstream device while simultaneously charging an internal battery of the wristband device.

Claim 15. A method of providing bidirectional neuromuscular functionality in a wristband device, the method comprising: providing a plurality of platinum electrodes on an inner surface of a wristband strap in contact with a user's skin; selectively configuring a multiplexer circuit to connect the plurality of platinum electrodes to a surface electromyography acquisition circuit to acquire muscle activity signals; and selectively configuring the multiplexer circuit to connect the plurality of platinum electrodes to a transcutaneous electrical nerve stimulation generation circuit to deliver therapeutic electrical stimulation.

Claim 16. The method of Claim 15, further comprising classifying the acquired muscle activity signals using a machine learning model to recognize hand gestures.

Claim 17. The method of Claim 15, further comprising generating biphasic symmetric TENS waveforms according to at least one of the following clinical protocols: conventional, acupuncture-like, burst, brief-intense, or hyperstimulation.

ABSTRACT

A modular wristband health monitoring apparatus comprises a zero-hole architecture in which a USB Type-C female receptacle and a USB Type-C male plug constitute the sole mechanical clasp and sole electrical interface of the device, with no additional housing openings. The USB-C female receptacle serves simultaneously as the breath analysis inlet, receiving exhaled breath through a Venturi-accelerated channel containing a PTFE hydrophobic membrane, electrochemical fuel cell for blood alcohol concentration measurement, and MOx VOC sensor for ketosis and biomarker detection. A flexible wristband strap incorporates six platinum electrodes selectively configurable by a multiplexer circuit to operate in a surface electromyography (sEMG) sensing mode for gesture recognition or a transcutaneous electrical nerve stimulation (TENS) therapy mode for pain management. Additional sensors include photoplethysmography, electrocardiography, skin temperature, UV index, and inertial measurement. A pass-through power delivery architecture allows simultaneous device charging and downstream device charging via the USB-C interface. The apparatus is controlled by a Nordic nRF52840 MCU with BLE 5.3 wireless connectivity.

Inventor: Srikanth Patchava Affiliation: Embedded Operating Systems Research Foundation (EoS Foundation) Date: May 2026 This application is filed by the inventor individually. The inventor retains full ownership of all rights disclosed herein.